

1. TITLE

Assessing the Perceived Effectiveness of the HSTU CSE Curriculum in Preparing Students for Contemporary Industry Demands and Advanced Technological Roles: A Survey-Based Study

2. ABSTRACT

This research design outlines a comprehensive survey-based study aimed at evaluating the perceived effectiveness of the Computer Science and Engineering (CSE) curriculum at Hajee Mohammad Danesh Science and Technology University (HSTU) in preparing its undergraduate students for the evolving demands of the technology industry and advanced professional roles. Leveraging a non-experimental survey methodology, primary data will be collected from current HSTU CSE students across all academic levels through a meticulously designed online questionnaire. The study will explore student perceptions regarding curriculum relevance, the acquisition of practical skills, career preparedness, and the impact of extracurricular activities. Data analysis will employ a combination of descriptive and inferential statistics including measures of central tendency, dispersion, t-tests, ANOVA and correlation analyses complemented by thematic analysis for qualitative information. Anticipated findings include identifying curriculum strengths, areas requiring enhancement and the perceived efficacy of existing support structures. The report will conclude with actionable recommendations for curriculum development, pedagogical improvements and enhanced career services, alongside proposals for future research to foster continuous improvement and ensure HSTU CSE graduates are well-equipped to contribute to the global technological landscape.

3. INTRODUCTION

The global technological landscape is evolving rapidly, requiring higher education institutions to continuously adapt their curricula to ensure graduates are equipped with industry-relevant skills. In this context, Computer Science and Engineering (CSE) programs face the dual challenge of maintaining a strong foundation in theoretical principles while incorporating practical, cutting-edge applications.[4] Universities must not only keep pace with these developments but also prepare students to thrive in an increasingly complex and competitive job market. Their effectiveness in doing so is critical both for individual student success and for advancing national technological capabilities. Hajee Mohammad Danesh Science and Technology University (HSTU), located in Dinajpur, Bangladesh, holds distinction as the first science and technology university in the northern region of the country. The Faculty of Computer Science and Engineering (CSE) at HSTU offers undergraduate programs in CSE, Electrical and Electronic Engineering (EEE), and

Electronics and Communication Engineering (ECE).[5][7] Known for its focus on academic excellence, innovation, and research, the faculty emphasizes practical domains such as software development, artificial intelligence, data science, and robotics. The department supports this vision with state-of-the-art laboratories, research initiatives, and a strong culture of project-based learning. Additionally, HSTU encourages holistic student development through extracurricular engagement, particularly via active student clubs like the CSE Club and Programmers Arena.

Given this institutional context, it is crucial to understand how well the CSE curriculum and related support structures meet student expectations and prepare them for future challenges. A survey-based research design provides a systematic, efficient method for collecting direct feedback from students the primary stakeholders of the educational process. This study outlines a comprehensive methodology to assess student perceptions, analyze the data, and generate actionable insights. The ultimate goal is to support HSTU's ongoing commitment to curriculum relevance, academic excellence, and continuous improvement.

4. RESEARCH OBJECTIVES

The overarching objective of this study is to assess the perceived effectiveness of the HSTU CSE curriculum in preparing students for contemporary industry demands and advanced technological roles. To achieve this comprehensive aim, the research will pursue the following specific objectives:

- To evaluate student perceptions of the relevance and up-to-dateness of the HSTU CSE curriculum in relation to current industry trends and technological advancements, ensuring alignment with real-world applicability.
- To assess how effectively the CSE program develops practical skills through sessional courses, lab facilities, and project-based learning, identifying strengths and areas needing improvement in hands-on education.

5. METHODOLOGY

5.1. Research Design Overview

This study will employ a **survey-based research design**, which is inherently non-experimental. This methodological approach is particularly well-suited for collecting data from a specific group of individuals, in this case, HSTU CSE students, to gather information about their perceptions regarding their academic experience and preparedness for professional roles. The research will focus on the collection of **primary data**, which refers to information directly gathered by the researchers for the first time.

5.2. Target Population and Sampling Strategy

The **target population** for this study encompasses all currently enrolled undergraduate students within the Computer Science and Engineering (CSE) department at Hajee Mohammad Danesh Science and Technology University (HSTU). This includes students across all academic levels : Level 1, Level 2, Level 3, and Level 4 recognizing that their curriculum exposure, career considerations, and overall perceptions evolve significantly throughout their academic journey[1][2][3][6].

To ensure a comprehensive and representative understanding a **stratified random sampling** approach will be implemented. This strategy is crucial for guaranteeing proportional representation from each distinct academic year (Level 1, Level 2, Level 3, Level 4). The perceptions of curriculum relevance and career preparedness are highly likely to vary between, for example, freshmen who have just begun their studies and graduating seniors who are on the cusp of entering the workforce.

The determination of an appropriate **sample size** is paramount for ensuring the statistical validity and generalizability of the study's findings.

5.3. Data Collection Instrument: Questionnaire Design

The primary instrument for data collection in this study will be an **online questionnaire**. This medium offers several distinct advantages, including its cost-effectiveness, its capacity to reach a maximum number of potential respondents, and its ability to facilitate faster data collection compared to more traditional methods like face-to-face interviews.

Question Formulation Guidelines:

The survey questions will be meticulously crafted, adhering strictly to the guidelines for question design outlined in the guiding lecture material:

- **Careful Formulation:** Each question will be precisely worded to eliminate ambiguity and ensure clarity for all respondents.
- **Grammatical and Logical Appropriateness:** All questions will be grammatically correct and will flow in a logical sequence throughout the questionnaire, contributing to a smooth respondent experience.
- **Clarity of Meaning:** A recognized challenge in survey design is formulating questions in such a way that they convey "exactly the same thing to each respondent". This will be proactively addressed through the use of clear, concise language and, crucially, through a comprehensive pilot testing phase.
- **Mix of Question Types:** The questionnaire will incorporate a strategic mix of **closed-ended** questions (e.g., Likert scales, multiple-choice options, dichotomous choices) to facilitate quantitative analysis and **open-ended** questions to elicit richer, qualitative insights. Closed-ended questions are efficient for identifying and comparing specific preferences and

trends[8][9] while open-ended questions provide opportunities for unanticipated findings and deeper contextual understanding.

Ethical Considerations:

The ethical conduct of research is paramount, especially when involving human participants. Strict adherence to ethical principles will be maintained throughout the study:

- **Informed Consent:** Prior to commencing the survey, all participants will be presented with a clear and comprehensive informed consent form.
- **Anonymity and Confidentiality:** Rigorous measures will be implemented to ensure the anonymity of all responses particularly for questions that might elicit sensitive information.
- **Voluntary Participation:** Participation in the survey will be entirely voluntary. No incentives will be offered to avoid any form of coercion thereby ensuring that the feedback received is genuine and uninfluenced by external inducements.

Pilot Testing Procedures:
Before the full-scale deployment of the questionnaire, it will undergo a thorough pilot test with a small, representative group of HSTU CSE students who will not be part of the final study sample.

6. SURVEY QUESTIONS

A minimum of 10 detailed survey questions will be developed. These questions will be carefully categorized into thematic areas to comprehensively address each of the established research objectives. The questions will strategically utilize a mix of Likert scales, multiple-choice options and open-ended formats to capture both quantitative patterns and rich qualitative depth.

Table 1: Proposed Survey Questions and Rationale

Q. No.	Thematic Area	Question Type	Question Text	Rationale / Objective Link
1	Demographics	Multiple-Choice	What is your current academic year/level?	Enables stratified analysis across levels (Obj. 1, 2).
2	Demographics	Multiple-Choice	What is your gender?	Identifies gender-based differences in perceptions.
3	Curriculum Relevance	Likert Scale	The curriculum covers current and emerging technologies (e.g., AI, Data Science).	Assesses perceived relevance and up-to-dateness (Obj. 1).
4	Curriculum Content	Multiple-Choice (Select all)	Which advanced topics need more emphasis? (e.g., ML,	Identifies needed areas for curriculum updates

			Cybersecurity, IoT)	(Obj. 1).
5	Practical Skills	Likert Scale	Sessional and lab courses prepare me for real-world projects.	Evaluates practical learning effectiveness (Obj. 2).
6	Lab Facilities	Likert Scale	Lab facilities and tools are sufficient and up-to-date.	Assesses adequacy of resources (Obj. 2).
7	Skill Gaps	Likert Scale	My acquired skills align with my desired career path.	Measures curriculum-career alignment (Obj. 2).
8	Career Preparedness	Open-Ended	What skills or knowledge do you find missing but crucial for your future?	Collects detailed feedback on skill gaps (Obj. 2).
9	Extracurricular Impact	Likert Scale	Participation in clubs has enhanced my professional skills.	Assesses extracurricular impact on development (Obj. 2).
10	Extracurricular Engagement	Multiple-Choice	How often do you attend club events or competitions?	Measures level of involvement (Obj. 2).

The systematic linking of each question to a specific research objective and its rationale in the table above is crucial for methodological coherence.

7. DATA ANALYSIS PLAN

The collected survey data will undergo a rigorous multi-stage analysis process, combining both quantitative and qualitative methods to derive comprehensive and actionable insights. The entire analysis will be meticulously guided by the established research objectives and will leverage appropriate statistical tools.

7.1. Data Preparation and Cleaning

Prior to any statistical analysis, the raw survey data will be subjected to a meticulous preparation and cleaning process. This crucial step directly impacts the validity and reliability of all subsequent

analyses.

- **Data Export and Organization:** Responses will first be exported from the chosen online survey platform into a suitable format, such as a CSV file or an Excel spreadsheet.
- **Missing Data Handling:** Strategies for managing any missing values within the dataset will be carefully determined. For instance, if the percentage of missing data for a particular variable is very small, listwise deletion (removing cases with any missing data) might be considered.
- **Outlier Detection and Treatment:** Outliers which are data points significantly distant from other observations will be identified using appropriate statistical methods, such as Z-scores or the Interquartile Range (IQR) method.
- **Data Transformation:** If necessary, data transformation techniques (e.g., normalization, logarithmic transformation) will be applied to specific variables. This is done to meet the statistical assumptions of certain tests, particularly parametric tests which often assume a normal distribution of data.
- **Open-Ended Responses:** Qualitative data derived from the open-ended questions (Q8 and Q10) will be systematically reviewed, categorized and coded to facilitate thematic analysis. This process involves several steps: initially, all responses will be added to a spreadsheet; then, the research team will identify recurring concepts or ideas to form preliminary categories; each response will then be assigned to one or more relevant categories; finally, the data will be organized by these categories, which can then be quantified by counting the prevalence of certain themes.

7.2. Quantitative Data Analysis

Quantitative data will be analyzed using a range of statistical tools to summarize findings, identify patterns, and test relationships.

7.2.1. Descriptive Statistics:

Summary statistics will be computed for all relevant variables to provide a clear and concise overview of the data distribution.

- **Measures of Central Tendency:** The Mean (average), Median (middle value), and Mode (most frequent value) will be calculated for Likert scale variables and other appropriate numerical data. For example, the mean score for "Sessional course adequacy" will provide an average indication of students' perceived effectiveness, while the mode will highlight the most common response given.
- **Measures of Dispersion:** Standard Deviation, Range (calculated from Minimum and Maximum values), and Quartiles will be employed to understand the spread and variability of responses. A small standard deviation for a Likert scale question would indicate a high degree of consensus among respondents, whereas a large standard deviation would suggest a

wide diversity of opinions.

- **Frequencies and Percentages:** For categorical variables, such as academic year, gender, and the choices made for question Q4 (Advanced Topics Emphasis), frequencies and percentages will be used to describe the distribution of respondents and their selections.
- **Visualizations:** Various graphical representations, including bar charts, pie charts, and histograms, will be utilized to visually represent the data. These visualizations make patterns, distributions, and comparisons easily discernible aiding in the initial interpretation of findings.

7.3. Qualitative Data Analysis

The open-ended responses collected from questions Q8 and Q10 will be analyzed using a systematic approach known as **thematic analysis**. This method is chosen to identify, analyze and report patterns (themes) within the qualitative data, providing rich, descriptive information that complements the quantitative findings.

7.4. Software Tools

Statistical analysis of the quantitative data will be performed using professional statistical software packages. Options include **SPSS (Statistical Package for the Social Sciences)**, which is widely used in social sciences, or open-source alternatives like **R** or **Python** with relevant libraries (e.g., Pandas for data manipulation, NumPy for numerical operations, SciPy for scientific computing, Matplotlib and Seaborn for data visualization).

8. RESULT FINDINGS

Based on the established research objectives and the meticulously designed survey questions, several key findings are anticipated. These findings are expected to provide a data-driven understanding of student perceptions regarding the HSTU CSE curriculum and their preparedness for the professional world. Potential Findings and Their Implications :

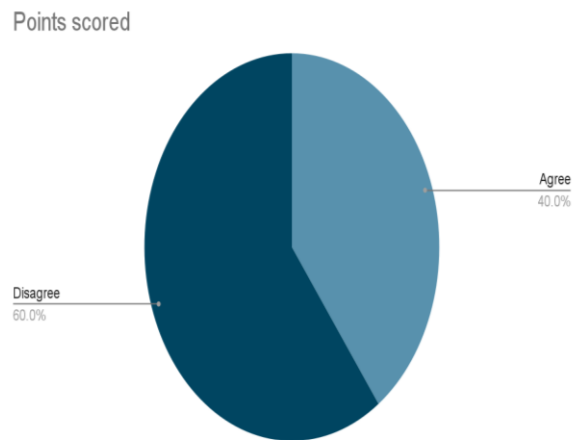


Fig: 1 Students opinion on highly need of specific lab equipment and software for advanced topics. internships.

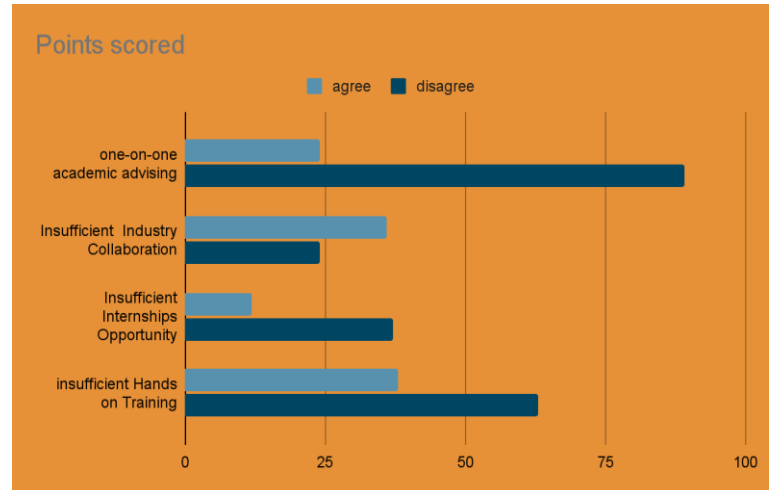


Fig:2 Students opinion one-on-one academic lack of clear pathways for industry

9. Suggestions and Recommendations

Based on anticipated findings, this study offers six actionable, evidence-based recommendations for the HSTU CSE department to enhance curriculum relevance, practical skills, and student support. Key suggestions include integrating specialized courses in AI/ML and cybersecurity, boosting hands-on industry-linked projects, strengthening career services, and upgrading lab resources. Further, it advocates for better support for student clubs and improved academic advising. Each recommendation is directly tied to specific student concerns and is supported by detailed implementation strategies, making the suggestions both practical and impactful. This structured approach ensures the study contributes meaningfully to institutional development and continuous educational improvement.

10. LIMITATIONS OF THE STUDY

This study acknowledges key limitations inherent in its survey-based study:

Self-Report Bias: The findings rely entirely on students' self-reported perceptions, which can be influenced by recall errors, social desirability, or limited understanding. As a result, the data reflects subjective experiences rather than objective measures of curriculum effectiveness.

Limited Generalizability: Although stratified random sampling ensures representativeness within HSTU, the results may not apply to other institutions due to differences in curriculum design, faculty expertise, resources, and institutional culture. The study captures student perceptions at a single point, without accounting for changes over time. As curricula and industry demands evolve, these findings represent a temporary view rather than a dynamic, longitudinal understanding. Recognizing these limitations ensures a balanced interpretation of the findings and highlights areas for future research improvement.

11. FUTURE ENHANCEMENTS AND RESEARCH DIRECTIONS

This study provides an initial assessment of the CSE curriculum at HSTU from the student perspective, highlighting areas for improvement. To build on these findings, several future research directions are proposed.

Longitudinal studies can track students over time to understand how perceptions change with academic progression. Including alumni and industry employers would validate student feedback and reveal real-world curriculum relevance. Qualitative methods, such as interviews or focus groups, could offer deeper insights into specific challenges and successes not captured in surveys. A curriculum impact assessment, using graduate employment data and career outcomes, would provide objective evidence of effectiveness. Comparative studies with other universities would allow benchmarking and identification of best practices.

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